FedFresh Proof (Compact)

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Notation

We write $w^t \in \mathbb{R}^d$ for the global model at round t, η_g for the server learning rate, T for the total rounds, and L, σ_ℓ^2, σ_g^2, G, γ for the smoothness and noise constants defined in Assumptions 1–5. Non-standard symbols specific to the system model are collected below.

Symbol	**Description**
Gradients and updates	
$g_u(w)$	Stochastic gradient of client u at w
$G^t = w^{t+1} - w^t$	Server update (delta) at round t
Δ^t	Scaled update: $G^t = -\eta_g \Delta^t$
X^t	Fresh estimator: $\frac{1}{\lvert S_t \rvert} \sum_k d(\tau_k)\, g_k(w^t)$
Client availability and staleness	
$\tau_u(t)$	Staleness of client u's update at round t
$d(\tau)$	Discount function applied to a gradient of staleness τ
$P_{\text{on}}^{(u)}$	Probability that client u is online
$R_u = N_u P_{\text{on}}^{(u)}$	Effective participation rate of client u
$\alpha_u = \mathbb{E}_{\tau_u \mid u}[d(\tau_u)]$	Mean discount under client u delay distribution
$\beta_u = \mathbb{E}_{\tau_u \mid u}[d^2(\tau_u)]$	Mean squared discount under client u delay distribution
Aggregate system quantities	
$K = \sum_u p_u R_u$	Expected number of participating clients per round
$C(\lambda) = \sum_u p_u R_u$	System throughput (total effective participation)
$\Gamma(\lambda) = \sum_u p_u \beta_u$	Aggregate second moment of discount weights
$N = \sum_u \lvert \frac{C(\lambda)}{K \lvert \mathcal{U} \rvert} - \frac{p_u R_u}{K} \rvert$	Participation imbalance; bias from heterogeneous rates
Derived convergence quantities	
$A(\lambda) = \gamma \big(\frac{C(\lambda)}{K} \big)^2$	Acceleration factor from gradient alignment
$H(\lambda) = 1 - \frac{2C(\lambda)}{K} + \frac{\Gamma(\lambda)}{K}$	Staleness-slack factor
$U(\lambda)$	Irreducible noise floor; see Theorem 1

Assumptions

Assumption 1 (Unbiased client gradient)**.** *$\mathbb{E}[g_u(w)] = \nabla F_u(w)$ for all $u \in \mathcal{U}$, $w \in \mathbb{R}^d$.*

Assumption 2 (Bounded variance)**.** *For all $u \in \mathcal{U}$, $w \in \mathbb{R}^d$:*

$$\mathbb{E}_{g_u|u}\big[\|g_u(w) - \nabla F_u(w)\|^2\big] \le \sigma_\ell, \qquad \|\nabla f(w) - \nabla F_u(w)\|^2 \le \sigma_g.$$

Assumption 3 (Bounded gradients)**.** *$\|\nabla F_u(w)\|^2 \le B$ for all $u \in \mathcal{U}$, $w \in \mathbb{R}^d$.*

Assumption 4 (L-smooth gradients)**.** *$\|\nabla F_u(w) - \nabla F_u(w')\|^2 \le L\|w - w'\|^2$ for all u, w, w'.*

Assumption 5 (Gradient alignment)**.** *There exists $\gamma \in (0,1]$ such that for any weights $W_u \ge 0$:*

$$\Big\|\sum_{u\in\mathcal{U}} W_u \nabla F(w^t)\Big\|^2 \;\ge\; \gamma\Big(\sum_{u\in\mathcal{U}} W_u\Big)^2 \|\nabla f(w^t)\|^2.$$

Supporting Lemmas

Lemma 1 (Second-moment bound)**.** *$\mathbb{E}\big[\|g_k(w)\|^2\big] \le 3(\sigma_\ell^2 + \sigma_g^2 + G)$.*

Proof. By the law of total expectation and three-term Young's inequality:

$$\begin{aligned}\mathbb{E}\big[\|g_k(w)\|^2\big] &= \mathbb{E}_k\,\mathbb{E}_{g|k}\big[\|g_k - \nabla F_k + \nabla F_k - \nabla f + \nabla f\|^2\big] \\ &\le 3\,\mathbb{E}_k\big[\mathbb{E}_{g|k}\big[\|g_k - \nabla F_k\|^2\big] + \|\nabla F_k - \nabla f\|^2 + \|\nabla f\|^2\big] = 3(\sigma_\ell^2 + \sigma_g^2 + G).\end{aligned}$$

□

Lemma 2 (A.S. convergence of gradient estimator)**.** *Let $f(\tau_u(t))$ satisfy $\mathbb{E}_{\tau_u|u}[f(\tau_u)] = \alpha_u$, $\mathbb{E}_{\tau_u|u}[f^2(\tau_u)] = \beta_u$, and let $K_n = \kappa n$. Then*

$$\lim_{|U|\to\infty} \frac{1}{|S_t|}\sum_{k\in S_t} f(\tau_k(t))\, g_k(w) \;\xrightarrow{\text{a.s.}}\; \frac{1}{K}\sum_{u\in\mathcal{U}} p_u \alpha_u \nabla F_u(w_u).$$

Proof. Define the numerator and denominator sequences

$$N_{|U|} = \frac{1}{n}\sum_{u=1}^{|U|} \mathbf{1}_{A_u}\mathbf{1}_{\pi_u} f(\tau_u) g_u(w^t), \qquad D_{|U|} = \frac{1}{n}\sum_{u=1}^{|U|} \mathbf{1}_{A_u}\mathbf{1}_{\pi_u}.$$

Their means are $\mathbb{E}[N_{|U|}] = \frac{1}{n}\sum_u p_u\alpha_u \nabla F_u(w^t)$ and $\mathbb{E}[D_{|U|}] = \kappa$. Kolmogorov's criterion is satisfied because

$$\sum_{u=1}^{\infty} \frac{\sigma_{N_u}^2}{u^2} \;\le\; \max_u(\beta_u)\cdot 3(\sigma_\ell^2 + \sigma_g^2 + G)\sum_{u=1}^{\infty}\frac{1}{u^2} = \frac{\pi^2}{6}\max_u(\beta_u)\cdot 3(\sigma_\ell^2 + \sigma_g^2 + G) < \infty,$$

$$\sum_{u=1}^{\infty} \frac{\sigma_{D_u}^2}{u^2} \;\le\; \sum_{u=1}^{\infty}\frac{1}{u^2} = \frac{\pi^2}{6} < \infty.$$

Hence $(N_{|U|}, D_{|U|}) \xrightarrow{\text{a.s.}} (N, D)$. Since $\kappa \neq 0$, the continuous mapping theorem with $g(x,y) = x/y$ gives $N_{|U|}/D_{|U|} \xrightarrow{\text{a.s.}} N/D = \frac{1}{K}\sum_u p_u R_u \nabla F_u(w_u)$. The result follows by identifying $N_{|U|}/D_{|U|}$ with the stated sum. □

Lemma 3 (Expected gradient estimator limit)**.** *Under the same conditions as Lemma 2 with $f(\tau_u) \le 1$:*

$$\lim_{|\mathcal{U}|\to\infty} \mathbb{E}\left[\frac{1}{|S_t|}\sum_{k\in S_t} f(\tau_k(t))\, g_k(w_k)\right] = \frac{1}{K}\sum_{u\in\mathcal{U}} p_u\alpha_u \nabla F_u(w_u).$$

Proof. We need to interchange expectation and limit. The ratio $D_{|U|}/N_{|U|}$ is dominated by

$$\mathbb{E}\left[\frac{1}{|S_t|}\sum_{k\in S_t}\|g_{u,k}(w^t)\|^2\right] \le 3(\sigma_\ell^2+\sigma_g^2+G) < \infty,$$

where the bound follows from Lemma 1. The Vitali convergence theorem therefore justifies the exchange, and the result follows from Lemma 2. □

Lemma 4 (Bias of gradient estimator)**.** *Define* $X^t = \frac{1}{|S_t|}\sum_{k\in S_t} d(\tau_k(t))\, g_k(w^t)$ *and* $N = \sum_{u\in\mathcal{U}}\left|\frac{C(\lambda)}{K|\mathcal{U}|} - \frac{p_u R_u}{K}\right|$. *Then*

$$\lim_{|\mathcal{U}|\to\infty} \mathbb{E}\big[\|\nabla f(w^t) - X^t\|^2\big] \;\le\; N^2\sigma_g.$$

Proof. Decompose $\|\nabla f(w^t) - X^t\|^2$ into a bias term $T_3 = \|\nabla f(w^t) - \mathbb{E}[X^t]\|^2$ and a variance term $T_4 = \|X^t - \mathbb{E}[X^t]\|^2$ via the bias–variance identity $\mathbb{E}\|Z-c\|^2 = \|\mathbb{E}[Z]-c\|^2 + \mathbb{E}\|Z-\mathbb{E}[Z]\|^2$.

Bounding T_3. Using $\sum_u\big(\frac{C}{|\mathcal{U}|} - \frac{p_u R_u}{K}\big) = 0$:

$$\lim_{|\mathcal{U}|\to\infty} T_3 = \left\|\sum_u\Big(\tfrac{C}{|\mathcal{U}|} - \tfrac{p_u R_u}{K}\Big)(\nabla F_u(w^t) - \nabla f(w^t))\right\|^2 \;\le\; N^2\sigma_g.$$

Bounding T_4. Let $\mu^t_{|\mathcal{U}|} = \mathbb{E}[G^t_{|\mathcal{U}|}]$ and $\mu = \frac{1}{K}\sum_u p_u^{(\mathrm{on})} p_u \alpha_u \nabla F_u(w_u)$. Expanding $\|G^t_{|\mathcal{U}|} - \mu^t_{|\mathcal{U}|}\|^2$ and noting $\mu^t_{|\mathcal{U}|} \to \mu$,

$$\lim_{|\mathcal{U}|\to\infty} T_4 = \lim_{|\mathcal{U}|\to\infty}\mathbb{E}\Big[\|G^t_{|\mathcal{U}|} - \mu\|^2\Big] = 0$$

by Lemma 2 and the continuous mapping theorem. Hence $\lim_{|\mathcal{U}|\to\infty}\mathbb{E}[T_1] = T_3 + T_4 \le N^2\sigma_g$. □

Lemma 5 (Staleness error bound)**.** *With* $C(\lambda) = \sum_u p_u R_u$, $\Gamma(\lambda) = \sum_u p_u\beta_u$, *and* $H(\lambda) = 1 - \frac{2C(\lambda)}{K} + \frac{\Gamma(\lambda)}{K}$*:*

$$\lim_{|\mathcal{U}|\to\infty}\left\|\frac{1}{|S_t|}\sum_{k\in S_t} d(\tau_k)\big(g_k(w^t) - g_k(w^{t-\tau_k+1})\big)\right\|^2 = \lim_{|\mathcal{U}|\to\infty}\mathbb{E}[T_2] \;\le\; \frac{6\sigma_\ell^2\,\Gamma(\lambda)}{K} + \frac{9\eta_g^2\,\Gamma(\lambda)\,L}{K}\,H(\lambda)\,(\sigma_\ell^2+\sigma_g^2+G).$$

Proof. **Step 1: Jensen and three-term expansion.** By Jensen's inequality,

$$\mathbb{E}[T_2] \;\le\; \mathbb{E}\left[\frac{1}{|S_t|}\sum_k d^2(\tau_k)\,\|g_k(w^t) - g_k(w^{t-\tau_k+1})\|^2\right].$$

Insert the decomposition $g_k(w^t) - g_k(w^{t-\tau_k+1}) = \big(g_k(w^t) - \nabla F_u(w^t)\big) + \big(\nabla F_u(w^t) - \nabla F_u(w^{t-\tau_u+1})\big) + \big(\nabla F_u(w^{t-\tau_u+1}) - g_k(w^{t-\tau_k+1})\big)$ and apply $(a+b+c)^2 \le 3(a^2+b^2+c^2)$. Taking expectations over g_k first:

$$\mathbb{E}[T_2] \le \mathbb{E}_{S_t}\left[\frac{3}{|S_t|}\sum_k \mathbb{E}_{\tau_k|S_t}\Big[d^2(\tau_k)\Big(2\sigma_\ell^2 + \mathbb{E}_{G^t|\tau_k,S_t}\big[\|\nabla F_u(w^t) - \nabla F_u(w^{t-\tau_u+1})\|^2\big]\Big)\Big]\right].$$

Step 2: Bound the gradient drift via L-smoothness. By Assumption 4,

$$\|\nabla F_u(w^t) - \nabla F_u(w^{t-\tau_u+1})\|^2 \;\le\; L\,\|w^t - w^{t-\tau_k+1}\|^2.$$

Writing $w^t - w^{t-\tau_k+1} = \sum_{\rho=t-\tau_k+2}^{t} G^{\rho-1}$ and applying Cauchy–Schwarz:

$$\|w^t - w^{t-\tau_k+1}\|^2 \;\le\; (\tau_k - 1)\sum_{\rho=t-\tau_k+2}^{t}\|G^{\rho-1}\|^2.$$

Hence,

$$\mathbb{E}[T_2] \;\le\; \frac{6\sigma_\ell^2}{K}\sum_u p_u\beta_u \;+\; \mathbb{E}_{S_t}\left[\frac{3L}{|S_t|}\sum_k \mathbb{E}_{\tau_k|S_t}\left[d^2(\tau_k)(\tau_k-1)\sum_\rho \mathbb{E}_{G^\rho}\big[\|G^{\rho-1}\|^2\big]\right]\right].$$

Step 3: Bound $\mathbb{E}[\|G^t\|^2]$ as $|\mathcal{U}| \to \infty$.

$$\begin{aligned}
\lim_{|\mathcal{U}|\to\infty} \mathbb{E}[\|G^t\|^2] &= \lim_{|\mathcal{U}|\to\infty} \mathbb{E}\left[\frac{\eta_g^2}{|S_t|}\sum_k d^2(\tau_k)\|g_k(w^{t-\tau_k})\|^2\right] \\
&\le \lim_{|\mathcal{U}|\to\infty} \mathbb{E}_{S_t}\left[\frac{\eta_g^2}{|S_t|}\sum_k \mathbb{E}_{\tau_k|S_t}\left[d^2(\tau_k)\right]\cdot 3(\sigma_\ell^2+\sigma_g^2+B)\right] \\
&= \frac{3\eta_g^2\,\Gamma(\lambda)}{K}(\sigma_\ell^2+\sigma_g^2+B),
\end{aligned}$$

where we used Lemma 1 and $\lim_{|\mathcal{U}|\to\infty}\frac{1}{K}\sum_u p_u\beta_u = \frac{\Gamma(\lambda)}{K}$.

Step 4: Collect terms and recover $H(\lambda)$. Substituting the Step 3 bound into the remainder term from Step 2 and summing over the $(\tau_k - 1)$ terms in $\sum_\rho$:

$$\mathbb{E}_{S_t}\left[\frac{3L}{|S_t|}\sum_k \mathbb{E}_{\tau_k}\left[d^2(\tau_k)(\tau_k-1)\sum_\rho \mathbb{E}[\|G^{\rho-1}\|^2]\right]\right] \;\le\; \frac{9\eta_g^2\Gamma(\lambda)L}{K}(\sigma_\ell^2+\sigma_g^2+B)\cdot\frac{1}{K}\sum_u p_u\,\mathbb{E}_\tau[d^2(\tau_u)(\tau_u-1)^2].$$

To evaluate $\mathbb{E}_\tau[d^2(\tau)(\tau-1)^2]$ in terms of $C(\lambda)$ and $\Gamma(\lambda)$, we specialise to $d(\tau)=1/\tau$, which gives $d^2(\tau)(\tau-1)^2 = (1-d(\tau))^2 = 1-2d(\tau)+d^2(\tau)$. Taking expectations and using the definitions of $C(\lambda)$ and $\Gamma(\lambda)$:

$$\frac{1}{K}\sum_u p_u\,\mathbb{E}_\tau[(1-d(\tau_u))^2] = 1-\frac{2C(\lambda)}{K}+\frac{\Gamma(\lambda)}{K} = H(\lambda).$$

Combining with the $\frac{6\sigma_\ell^2\Gamma(\lambda)}{K}$ term from Step 2 gives:

$$\lim_{|\mathcal{U}|\to\infty}\mathbb{E}[T_2] \;\le\; \frac{6\sigma_\ell^2\,\Gamma(\lambda)}{K} + \frac{9\eta_g^2\,\Gamma(\lambda)\,L}{K}\,H(\lambda)\,(\sigma_\ell^2+\sigma_g^2+B). \qquad \square$$

Main Theorem

Define the following quantities:

Theorem 1 (Convergence of FedFresh)**.** *For $K_{|\mathcal{U}|} = \kappa|\mathcal{U}|$,*

$$\lim_{|\mathcal{U}|\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(w^t)\|^2 \;\le\; \frac{2(f(w^0)-f(w^*))}{\eta_g T\,(1+(1-L\eta_g)A(\lambda))} + \frac{U(\lambda)}{1+(1-L\eta_g)A(\lambda)}.$$

Where,

$$U(\lambda) = N^2\sigma_g + \frac{9\eta_g^2\Gamma(\lambda)L}{K}\left(\frac{2\sigma_\ell^2}{3L\eta_g^2} + H(\lambda)(\sigma_\ell^2+\sigma_g^2+B)\right).$$

Proof. **Step 1: One-step descent.** From L-smoothness and the identity $\langle a,b\rangle = \frac{1}{2}(\|a\|^2+\|b\|^2-\|a-b\|^2)$:

$$\|\nabla f(w^t)\|^2 \;\le\; \frac{2}{\eta_g}(f(w^t)-\mathbb{E}[f(w^{t+1})]) + \mathbb{E}\left[\|\nabla f(w^t)-\Delta^t\|^2\right] - (1-L\eta_g)\,\mathbb{E}\left[\|\Delta^t\|^2\right], \tag{1}$$

where $\Delta^t = -\frac{1}{\eta_g}G^t = \frac{1}{|S_t|}\sum_k d(\tau_k)g_k(w^{t-\tau_k+1})$.

Step 2: Gradient estimator error. Decompose the error term:

$$\|\nabla f(w^t)-G^t\|^2 \;\le\; 2\underbrace{\|\nabla f(w^t)-X^t\|^2}_{T_1} + 2\underbrace{\|X^t_{\text{stale}}-X^t\|^2}_{T_2}.$$

Lemma 4 gives $\lim_{|\mathcal{U}|\to\infty}\mathbb{E}[T_1]\le N^2\sigma_g$, and Lemma 5 bounds $\lim_{|\mathcal{U}|\to\infty}\mathbb{E}[T_2]$.

Step 3: Acceleration via gradient alignment. By Assumption 5 and Jensen's inequality:

$$\mathbb{E}[\|\Delta^t\|^2] \;\geq\; \|\mathbb{E}[\Delta^t]\|^2 \;\geq\; \gamma\left(\frac{C(\lambda)}{K}\right)^2 \|\nabla f(w^t)\|^2 = A(\lambda)\|\nabla f(w^t)\|^2.$$

Step 4: Combine and telescope. Substituting the bounds from Steps 2–3 into (1), rearranging to collect $\|\nabla f(w^t)\|^2$ on the left, and dividing by $(1 + (1 - L\eta_g)A(\lambda))$ gives

$$\|\nabla f(w^t)\|^2 \;\leq\; \frac{2}{\eta_g(1 + (1 - L\eta_g)A(\lambda))}(f(w^t) - \mathbb{E}[f(w^{t+1})]) + \frac{U(\lambda)}{1 + (1 - L\eta_g)A(\lambda)}.$$

Summing over $t = 0, \ldots, T - 1$ and dividing by T telescopes the first term to $f(w^0) - f(w^*)$, completing the proof. □